B.Tech. Final Year Project (2019-2023)

Zeroth Review Report

Lattice Based Cryptanalysis

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Guide

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Scope of the Project

The development of quantum computation has raised concerns about the security of traditional cryptographic techniques such as RSA, ECC etc. Lattice based cryptography is a proposed solution to this problem.

The security of lattice-based cryptographic schemes heavily depends on the ability of an attacker to find the shortest vector in the underlying lattice.

The objectives of this project are:

- To analyze, identify and improve design issues related to efficiency. Study of lattice-based encryption schemes, identifying vulnerabilities and possible optimization of existing algorithms.
- Evaluation of various lattice-based cryptographic schemes and pinpointing any weaknesses and vulnerabilities.

The project aims to enhance the efficiency of lattice-based cryptography for practical use.

References

[1] Deng, X. (2016). An Introduction to Lenstra-Lenstra-Lovasz Lattice Basis Reduction Algorithm.

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[3] Mariano, Artur & Laarhoven, Thijs & Correia, Fabio & Rodrigues, Manuel & Falcao, Gabriel. (2017). A Practical View of the State-of-the-Art of Lattice-Based Cryptanalysis. IEEE Access. PP. 1-1. 10.1109/ACCESS.2017.2748179.



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